

SCO Approach to k-MDB

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1 Introduction

2 Definitions

Define a bipartite graph as a graph $G = (U, V, E)$, where U and V are two disjoint sets of vertices, and $E \subseteq U \times V$ is the set of edges between U and V . Let $S = (U_S, V_S)$ denote a subset of vertices where $U_S \subseteq U$ and $V_S \subseteq V$. Let $G(S) = (U_S, V_S, E_S)$ be the induced subgraph of S where $E_S = \{(u, v) \mid u \in U_S \wedge v \in V_S \wedge (u, v) \in E\}$. Define $\bar{E}(S) = \{(u, v) \mid u \in U_S \wedge v \in V_S \wedge (u, v) \notin E\}$ be the set of non-edges in a subgraph. Let a k -defective biclique be a subgraph S where $|\bar{E}(S)| \leq k$.

Define the neighbors of a node u in a subgraph S as $N_S(u) = \{v \mid \{u, v\} \in E(S)\}$. Define the nonneighbors of a node as $\bar{N}_S(u) = \{v \mid (u, v) \notin E(S)\}$. Define the degree of a node as $d_S(u) = |N_S(u)|$, and the nondegree as $\bar{d}_S(u) = |\bar{N}_S(u)|$.

Define expanding a k -MDB S as adding a vertex u such that $|\bar{E}(S \cup \{u\})| \leq k$. Define a maximal k -MDB as a subgraph S that cannot be expanded. That is, for any node $u \notin S$, $|\bar{E}(S \cup \{u\})| > k$. Define the maximum k -MDB as the largest maximal k -MDB in a graph G .

3 Methods

We propose Slime Colony Optimization (SCO), an adaptation of ACO fitted to biclique graphs. Pheromones allow agents to converge on optimal solutions, and subspecies of slimes allow for multiple near-optimal solutions.

Pseudocode:

Algorithm 1 SCO

Ensure: a subset of vertices S such that $G(S)$ is a k -MDB

```
function SCO( $G, N_a, N_g, E$ )  
  for all  $u \in G$  do  
     $P(u) \leftarrow 1$   
   $a \leftarrow 0$   
  while  $a < N_g$  do  
     $a \leftarrow a + 1$   
     $A \leftarrow N_a$  empty subgraphs  
    for all  $ant \in A$  do  
      six seven  
  for all  $u \in G$  do  
     $P(u) \leftarrow P(u) \times E$   
  return  $T$ 
```

NEXT: Figure out what datasets I want to compare against, implement subspecies, and then write the pseudocode